Further Math with Ali Hashir

# Rational Functions

Further Pure Mathematics 1 – Chapter 2 • CIE 9231

## About this chapter

A rational function is one polynomial divided by another. Almost everything about its graph is decided by two things: where the denominator vanishes, and what the function does when $|x|$ is large. Read those off the algebra and an accurate sketch follows without plotting a single point.

Two techniques carry the chapter. *Division*: rewriting $\dfrac{x^2+3x-1}{x-2}$ as $x+5+\dfrac{9}{x-2}$ shows at once that the curve closes in on the line $y=x+5$ far from the origin, because the remainder term dies away. *The discriminant*: fix $y$, rearrange into a quadratic in $x$, and demand a real solution — this says exactly which values $y$ can take, and the ends of a forbidden interval are the turning points, found without differentiating.

## Key results you will need

**1. Asymptotes** of $y=\mathrm{N}(x)/\mathrm{D}(x)$, where N and D have no common factor:

- **vertical** wherever $\mathrm{D}(x)=0$;
- $\deg \mathrm{N} < \deg \mathrm{D}$: horizontal asymptote $y=0$;
- $\deg \mathrm{N} = \deg \mathrm{D}$: horizontal asymptote $y=$ ratio of the leading coefficients;
- $\deg \mathrm{N} = \deg \mathrm{D}+1$: oblique asymptote $y=$ the quotient on division.

**2. The range, and the turning points.** Multiply up and collect as a quadratic in $x$ whose coefficients involve $y$. A point exists only if that quadratic has a real root, so discriminant $\geqslant 0$ — an inequality in $y$ alone. Any $y$ making the coefficient of $x^2$ vanish must be checked separately, as the equation is then linear. At a turning point the discriminant is *zero*, so putting each boundary value of $y$ back gives a repeated root — the $x$-coordinate, with no differentiation.

**3. Related curves**, each read off the graph of $y=\mathrm{f}(x)$:

- $y=\frac{1}{\mathrm{f}(x)}$: zeros $\leftrightarrow$ vertical asymptotes, $y=k$ becomes $y=\frac{1}{k}$, the sign is unchanged;
- $y^2=\mathrm{f}(x)$: exists only where $\mathrm{f}(x)\geqslant 0$, symmetrical in the $x$-axis;
- $y=\left|\mathrm{f}(x)\right|$: reflect in the $x$-axis whatever lies below it;
- $y=\mathrm{f}\left(|x|\right)$: keep $x\geqslant 0$ and reflect that part in the $y$-axis.

## How to use this worksheet

- The fifteen questions are in **increasing order of difficulty**: 1–5 rehearse asymptotes, 6–10 build ranges, turning points and related graphs, 11–15 are examination-length. Work through them in order.
- Marks are in square brackets. Allow about **$1\frac{1}{2}$ minutes per mark**; the full worksheet is roughly **$3\frac{1}{2}$ hours** of work.
- **Every sketch must show** the asymptotes (dashed), the intersections with the axes, any turning points, and the correct behaviour on *each* side of *each* asymptote. Shape and position matter; exact scale does not.
- Use the discriminant for any question asking you to show that $y$ lies in some set of values. Differentiate only when asked, or when there is no alternative.
- Leave every answer **exact**, and give inequalities as a set of values of $x$.
- Where a question says *show that*, all the marks are in the working.

**QUESTIONS 1–5** • Reading asymptotes off the algebra

## Question 1

State the equations of the asymptotes of each of the following curves.

**(a)** $y = \dfrac{3}{x-4}$ [2]

**(b)** $y = \dfrac{2x+1}{x-5}$ [2]

**(c)** $y = \dfrac{5}{x^2-9}$ [3]

## Question 2

Find the number of vertical asymptotes of each of the following curves, giving a reason in each case.

**(a)** $y = \dfrac{1}{x^2-7x+12}$ [2]

**(b)** $y = \dfrac{1}{x^2+4x+7}$ [2]

**(c)** $y = \dfrac{1}{x^2-6x+9}$ [2]

## Question 3

The curve $C$ has equation

$$y = \frac{x^2+3x-1}{x-2}.$$

**(a)** Express $y$ in the form $x + a + \dfrac{b}{x-2}$, where $a$ and $b$ are constants to be found. [3]

**(b)** Hence write down the equations of the two asymptotes of $C$. [2]

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## Question 4

The curve $C$ has equation

$$y = \frac{3x^2 - x + 2}{x^2 + x - 6}.$$

**(a)** Find the equations of the three asymptotes of $C$. [3]

**(b)** Show that $C$ does not meet the $x$-axis, and find the coordinates of the point where $C$ meets the $y$-axis. [3]

## Question 5

The function f is given by $\mathrm{f}(x) = x^2 - 4x + 3$.

**(a)** Write down the equations of the asymptotes of the curve $y = \dfrac{1}{\mathrm{f}(x)}$. [3]

**(b)** Find the coordinates of the turning point of $y = \dfrac{1}{\mathrm{f}(x)}$. [3]

**(c)** Sketch $y = \dfrac{1}{\mathrm{f}(x)}$, showing the asymptotes, the turning point and the point where the curve meets the $y$-axis. [3]

**QUESTIONS 6–10** • Ranges, related graphs and inequalities

## Question 6

For the curve $y = \dfrac{x-1}{x+2}$, determine the set of values of $x$ for which $y < 3$.

[4]

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## Question 7

The function f is given by $\mathrm{f}(x) = \dfrac{x-3}{x+1}$.

**(a)** Sketch $y = \mathrm{f}(x)$, showing the asymptotes and the intersections with the axes. [3]

**(b)** Sketch $y = \left|\mathrm{f}(x)\right|$. [2]

**(c)** Sketch $y = \mathrm{f}\left(|x|\right)$, and explain why this curve has no vertical asymptote. [4]

## Question 8

The curve $C$ has equation

$$y = \frac{x^2 + 4x - 5}{x - 3}.$$

**(a)** Express $y$ in the form $x + p + \dfrac{q}{x-3}$, and hence state the equations of the asymptotes of $C$. [4]

**(b)** Show that there is no point on $C$ for which $2 < y < 18$. [4]

**(c)** Find the coordinates of the turning points of $C$. [3]

## Question 9

Find the set of values of $x$ for which

$$\frac{3}{x-1} < x + 1.$$

[5]

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## Question 10

The curve $C$ has equation

$$y = \frac{x+1}{x^2-4}.$$

**(a)** Find the equations of the asymptotes of $C$. [3]

**(b)** Show that $C$ has no stationary points. [3]

**(c)** Sketch $C$, stating the coordinates of the intersections with the axes. [4]

**(d)** Find the set of values of $x$ for which $y > 0$. [2]

**QUESTIONS 11–15** • Examination-standard problems

## Question 11

The curve $C$ has equation

$$y = \frac{x^2+ax+b}{x-2},$$

where $a$ and $b$ are constants. One of the asymptotes of $C$ is $y = x+5$, and $C$ passes through the point $(0, 3)$.

**(a)** Find the values of $a$ and $b$. [4]

**(b)** State the equation of the other asymptote. [1]

**(c)** Find the coordinates of the turning points of $C$. [4]

## Question 12

The curve $C$ has equation

$$y^2 = \frac{x-2}{x+1}.$$

**(a)** Find the set of values of $x$ for which $C$ is defined. [3]

**(b)** State the equations of the three asymptotes of $C$. [2]

**(c)** Sketch $C$. [3]

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## Question 13

The curve $C$ has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = \frac{x^2 + 4x + 1}{x^2 + 1}.$$

**(a)** Show that $C$ has no vertical asymptote, and state the equation of the horizontal asymptote of $C$. [2]

**(b)** Show that $-1 \leqslant y \leqslant 3$ for all real values of $x$. [4]

**(c)** Find the coordinates of the turning points of $C$. [3]

**(d)** Sketch $C$, stating the coordinates of the intersections with the axes. [3]

**(e)** State the set of values of $k$ for which the equation $\mathrm{f}(x) = k$ has exactly two distinct real solutions. [3]

## Question 14

The curve $C$ has equation

$$y = \frac{x^2 + 2x - 3}{x^2 + 4x + 4}.$$

**(a)** State the equations of the asymptotes of $C$. [2]

**(b)** Find the coordinates of the points where $C$ meets the coordinate axes. [3]

**(c)** Show that $y \leqslant \frac{4}{3}$ at all points of $C$, and find the coordinates of the turning point of $C$. [6]

**(d)** Sketch $C$. [3]

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## Question 15

The curve $C$ has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = \frac{x^2 + 3}{x - 1}.$$

**(a)** Find the equations of the asymptotes of $C$. [3]

**(b)** Show that there is no point on $C$ for which $-2 < y < 6$. [4]

**(c)** Find the coordinates of the turning points of $C$. [3]

**(d)** Sketch $C$, stating the coordinates of the intersection with the $y$-axis. [3]

**(e)** Sketch, on a separate diagram, the curve $y = \left|\mathrm{f}(x)\right|$. [2]

**(f)** Hence find, in exact form, the set of values of $x$ for which $\left|\mathrm{f}(x)\right| < 6$. [4]

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Answers and fully worked solutions are available in the companion booklets.

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